

Chapter 7: Applications of Integration

Section 7.4: Arc Length and Surfaces of Revolution

Arc Length of a Curve

The objective is to find the length of a curve that has a starting and end points.

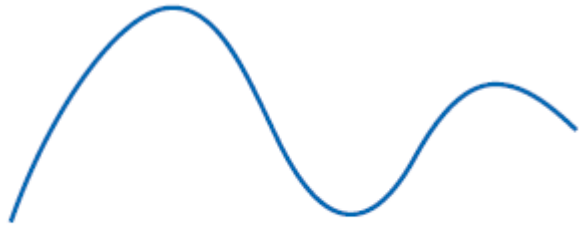


FIGURE 1

Consider a curve C that is defined by the equation $y = f(x)$, where f is continuous and $a \le x \le b$. See figure 1.

We apply the idea of the Riemann sums by dividing the interval into subintervals of equal width $\Delta x = \frac{b-a}{n}$.

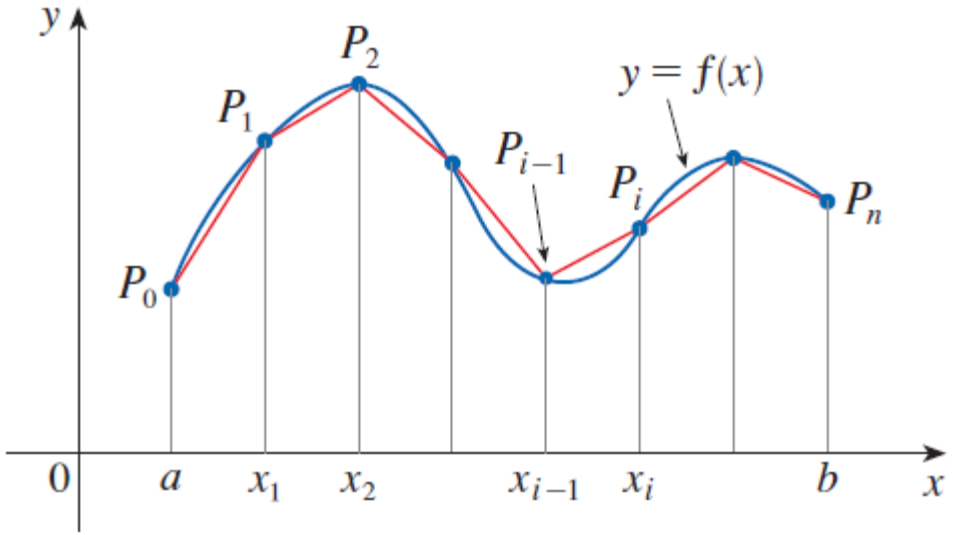


FIGURE 3

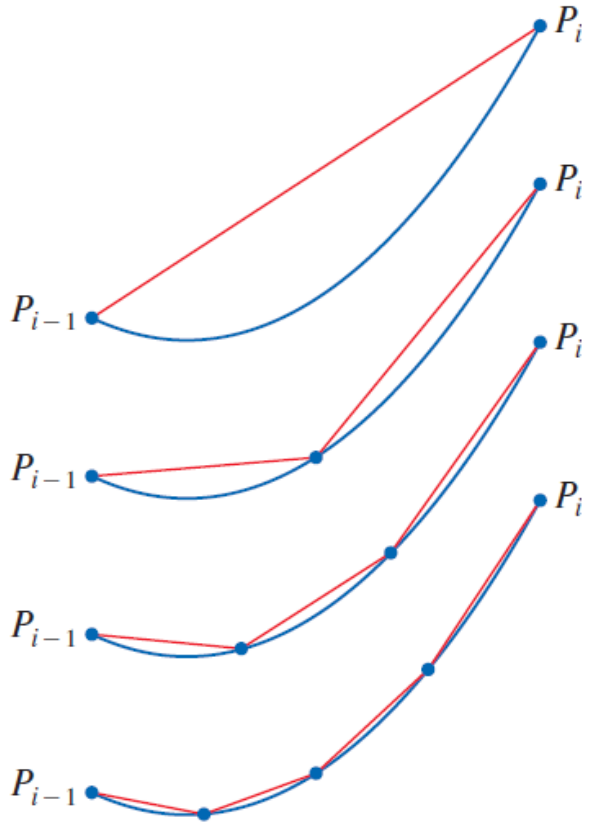


FIGURE 4

The length L of C is approximately the length of this polygonal path and the approximation gets better as we let n increase.

Arc Length of a Curve

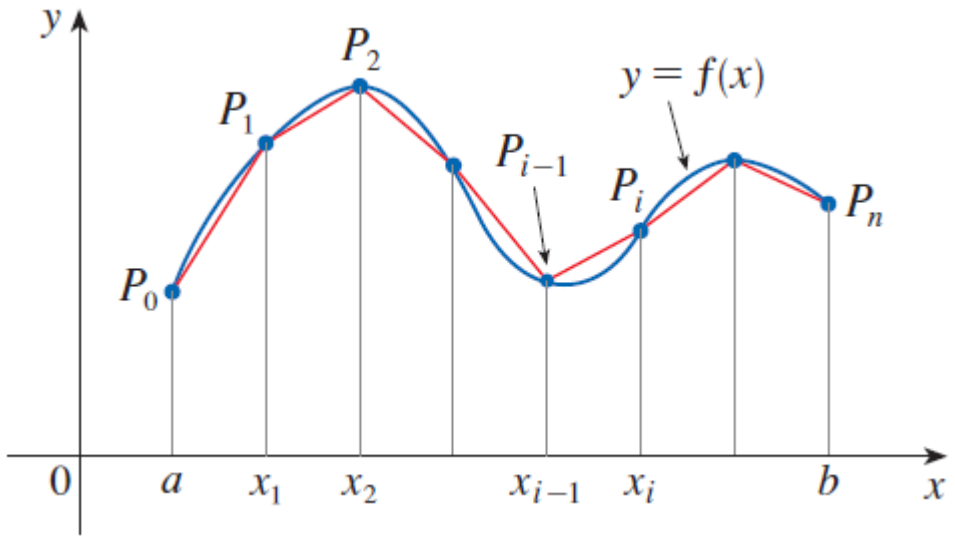


FIGURE 3

Therefore, we define the length L of the curve C with equation $y = f(x), a \le x \le b$, as the limit of lengths of these approximating polygonal paths (if the limit exists):

$$L = \lim_{n \to \infty} \sum_{i=1}^{n} |P_{i-1}P_i|$$

where $|P_{i-1}P_i|$ is the distance between the points P_{i-1} and P_i.

Arc Length of a Curve

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where $|P_{i-1}P_i|$ is the distance between the points P_{i-1} and P_i.

If we let $\Delta y_i = y_i - y_{i-1}$, then

$$|P_{i-1}P_i| = \sqrt{(x_i - x_{i-1})^2 + (y_i - y_{i-1})^2} = \sqrt{(\Delta x)^2 + (\Delta y_i)^2}$$

By applying the Mean Value Theorem to f on the interval $[x_{i-1}, x_i]$, we find that there is a number x_i^* between x_{i-1} and x_i such that

$$f(x_i) - f(x_{i-1}) = f'(x_i^*)(x_i - x_{i-1})$$

that is,

$$\Delta y_i = f'(x_i^*)\,\Delta x$$

Thus we have

$$\begin{aligned} |P_{i-1}P_i| &= \sqrt{(\Delta x)^2 + (\Delta y_i)^2} = \sqrt{(\Delta x)^2 + [f'(x_i^*)\,\Delta x]^2} \\ &= \sqrt{1 + [f'(x_i^*)]^2}\,\sqrt{(\Delta x)^2} = \sqrt{1 + [f'(x_i^*)]^2}\,\Delta x \qquad (\text{since } \Delta x > 0) \end{aligned}$$

Arc Length of a Curve

$$L = \lim_{n \to \infty} \sum_{i=1}^{n} |P_{i-1}P_i|$$

Thus we have

$$|P_{i-1}P_i| = \sqrt{(\Delta x)^2 + (\Delta y_i)^2} = \sqrt{(\Delta x)^2 + [f'(x_i^*)\,\Delta x]^2}$$
$$= \sqrt{1 + [f'(x_i^*)]^2}\,\sqrt{(\Delta x)^2} = \sqrt{1 + [f'(x_i^*)]^2}\,\Delta x \qquad (\text{since } \Delta x > 0)$$

Therefore, by Definition 1,

$$L = \lim_{n \to \infty} \sum_{i=1}^{n} |P_{i-1}P_i| = \lim_{n \to \infty} \sum_{i=1}^{n} \sqrt{1 + [f'(x_i^*)]^2}\,\Delta x$$

We recognize this expression as being equal to

$$\int_a^b \sqrt{1 + [f'(x)]^2}\,dx$$

by the definition of a definite integral. We know that this integral exists because the function $g(x) = \sqrt{1 + [f'(x)]^2}$ is continuous. Thus we have proved the following theorem:

Arc Length of a Curve

2 The Arc Length Formula If f' is continuous on $[a, b]$, then the length of the curve $y = f(x)$, $a \leqslant x \leqslant b$, is

$$L = \int_a^b \sqrt{1 + [f'(x)]^2}\, dx$$

If we use Leibniz notation for derivatives, we can write the arc length formula as follows:

3

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$

If a curve has the equation $x = g(y)$, $c \leqslant y \leqslant d$, and $g'(y)$ is continuous, then by interchanging the roles of x and y in Formula 2 or Equation 3, we obtain the following formula for its length:

4

$$L = \int_c^d \sqrt{1 + [g'(y)]^2}\, dy = \int_c^d \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy$$

EXAMPLE

Find the length of the curve $y = \frac{1}{3}\sqrt{x}\,(x-3), 1 \le x \le 9.$

SOLUTION

$$L = \frac{32}{3}$$

EXAMPLE

Find the length of the curve $x = \frac{1}{3}(y^2 + 2)^{\frac{3}{2}}, \quad 0 \leq y \leq 4.$

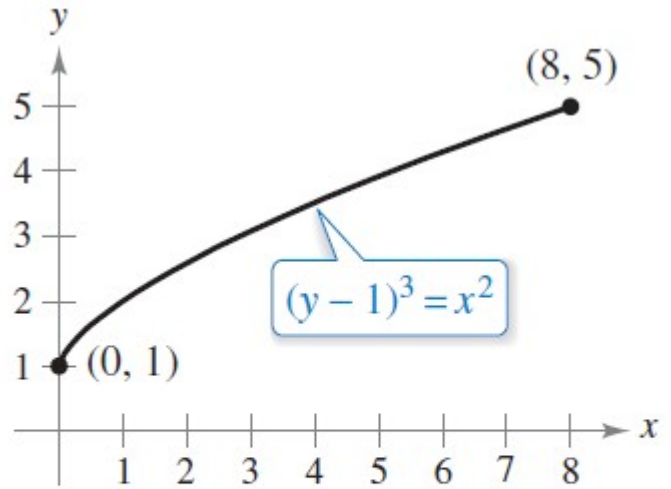
SOLUTION

$$L = \frac{76}{3}$$

EXAMPLE

Find the arc length of the graph of $(y - 1)^3 = x^2$ on the interval $[0, 8]$, as shown in Figure 7.40.

SOLUTION



The arc length of the graph of y on $[0, 8]$

Figure 7.40

$$L = \frac{1}{27}\left[\, 40^{\frac{3}{2}} - 4^{\frac{3}{2}}\right].$$

Area of a Surface of Revolution

A surface of revolution is formed when a curve is rotated about a line. Such a surface is the lateral boundary of a solid of revolution of the type discussed in Sections 7.2 and 7.3.

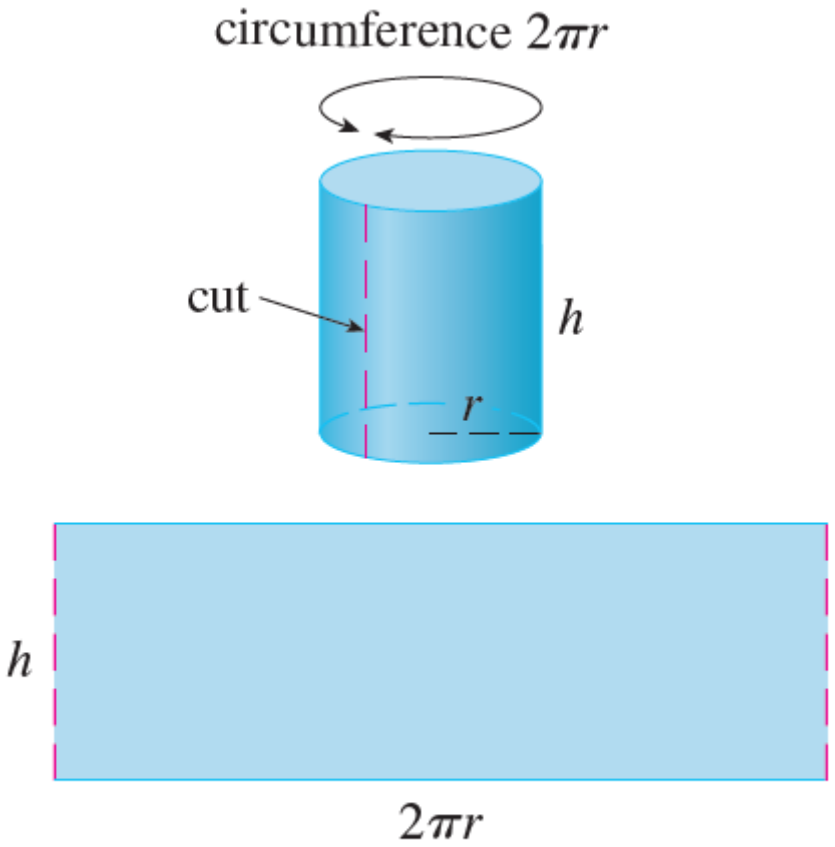


FIGURE 1

The lateral surface area of a cone is $A = \pi r l$.

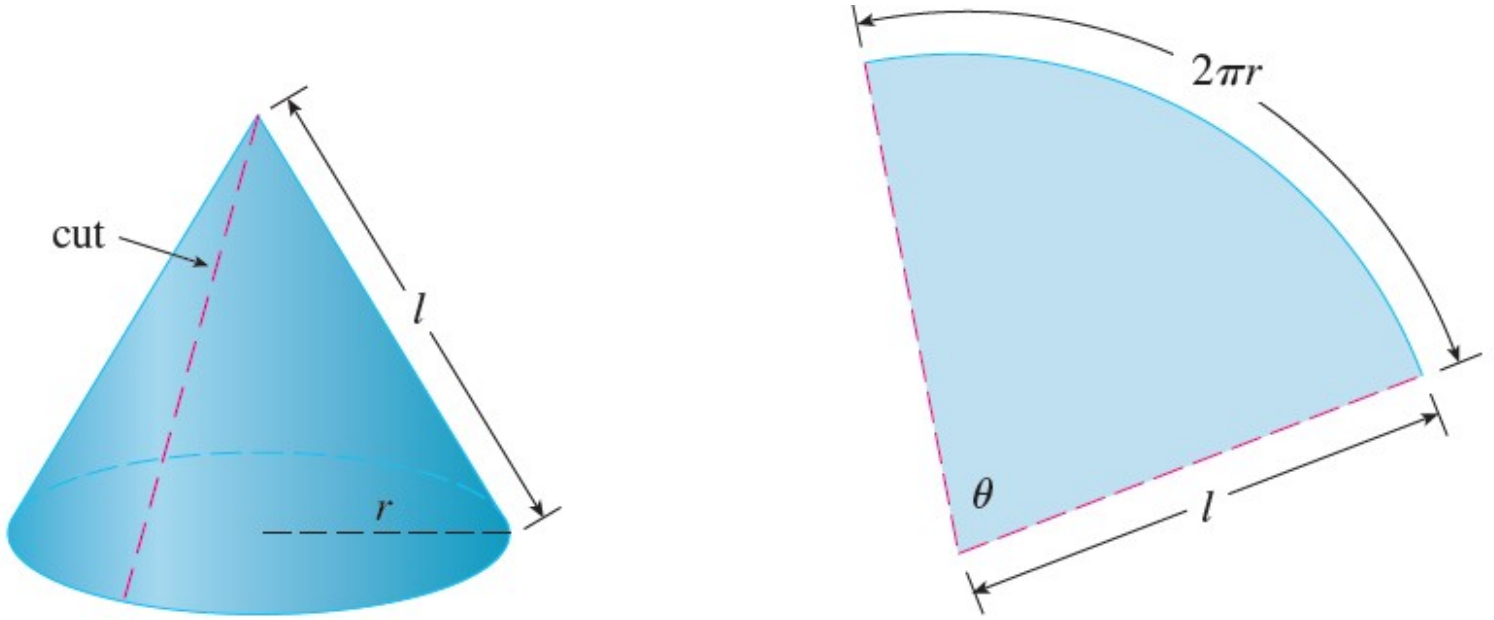


FIGURE 2

Read the textbook for deducing the surface area formulas.

Definition:

(1) If the function f is positive and f' is continuous on the interval $[a, b]$, then the area of the surface (or surface area) obtained by rotating the curve $y = f(x)$, $a \le x \le b$, about **the *x*-axis** is

$$S = \int_a^b 2\pi f(x)\sqrt{1 + [f'(x)]^2}\, dx$$

With the Leibniz notation for derivatives, this formula becomes

$$S = \int_a^b 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$

This is better formula to be remembered

(2) If the function f is positive and f' is continuous on the interval $[a, b]$, then the area of the surface (or surface area) obtained by rotating the curve $y = f(x)$, $a \le x \le b$, about **the *y*-axis** is

$$S = \int_a^b 2\pi x \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$

Read the textbook for deducing the surface area formulas.

Definition:

(3) If the function f is positive and f' is continuous on the interval $[a, b]$, then the area of the surface (or surface area) obtained by rotating the curve $x = g(y)$, $c \le y \le d$, about **the x-axis** is

$$S = \int_c^d 2\pi y \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy$$

(4) If the function f is positive and f' is continuous on the interval $[a, b]$, then the area of the surface (or surface area) obtained by rotating the curve $x = g(y)$, $c \le y \le d$, about **the y-axis** is

$$S = \int_c^d 2\pi x \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy$$

The above formulas can be summarized symbolically, using the notation for arc length given earlier, as

For rotation about x-axis	For rotation about y-axis	For rotation about x-axis	For rotation about y-axis
$S = \int 2\pi y \, ds$		$S = \int 2\pi x \, ds$	

where

$$ds = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx \qquad \text{or} \qquad ds = \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy$$

If the curve's equation is written as a function of x

If the curve's equation is written as a function of y

EXAMPLE

Find the area of the surface obtained by rotating the curve $x = \frac{y^2}{4} - \frac{\ln y}{2}, 1 \leq y \leq e$, about the x-axis.

SOLUTION

$$S = \frac{\pi}{3}(e^3 + 3e - 4)$$

EXAMPLE

Find the area of the surface obtained by rotating the curve $x = y^3, 1 \leq y \leq 2$, about the y-axis.

SOLUTION

$$S = \frac{\pi}{27}\left(145\sqrt{145} - 10\sqrt{10}\right)$$

Solve exercises of section 7.4.

Start with the suggested problems mentioned in the syllabus.